

Predicting UAV Flight Path Deviation from Onboard Telemetry and Wind Conditions Using Interpretable Machine Learning

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ABSTRACT

Unmanned aerial vehicles flying pre-planned survey patterns drift off their intended track whenever wind overpowers the autopilot’s ability to compensate, degrading the quality of downstream products such as orthomosaics and soil-moisture maps. We study cross-track deviation on 15 real outdoor flights of a multi-rotor UAV flown by Dr. Hemanth Dakshinamurthy’s group at SC State University, framed as both instantaneous regression (deviation in metres) and high-deviation classification (75th-percentile threshold). We compare three interpretable course-covered models (decision tree, k -nearest neighbors, random forest) across *three* families of features: hourly public weather (Open-Meteo reanalysis), onboard ArduPilot telemetry (attitude, motor output, rates, vibration, EKF innovations), and the union of the two. Evaluated under 5-fold GroupKFold cross-validation that holds out entire flights, onboard telemetry yields the best regression ($R^2 = 0.52$) and the best classification ($F_1 = 0.89$) whereas hourly public weather alone gives $R^2 < 0$ — actively worse than a mean predictor. Feature-importance analysis identifies altitude, EKF horizontal innovation, and ground speed as the dominant onboard predictors, consistent with the view that a multi-rotor’s own flight-state telemetry is a better high-frequency wind-response sensor than any off-board anemometer we could realistically access. We release the pipeline at <https://github.com/iiahakk/-uav-flight-path-deviation-ml>.

KEYWORDS

unmanned aerial vehicles, flight path deviation, ArduPilot telemetry, decision trees, random forest, feature engineering, GroupKFold

1 INTRODUCTION AND BACKGROUND

1.1 Problem

Unmanned aerial vehicles (UAVs) are now routinely used for precision agriculture, disaster surveying, and hyperspectral remote sensing, where strict adherence to a planned flight path is critical to the quality of downstream products such as orthomosaics, soil-moisture maps, and canopy reflectance estimates. A persistent challenge is wind: even moderate gusts cause measurable cross-track drift between the drone’s recorded GPS trajectory and the planned waypoint sequence, reducing image overlap, introducing pixel-alignment artifacts between passes, and in the worst case forcing entire missions to be re-flown. The general problem we address is: *given a set of recorded outdoor UAV survey flights with both the desired waypoint polyline and rich onboard telemetry, can*

we build interpretable machine-learning models that predict cross-track deviation, and can we identify which features drive the high-deviation tail? We refine the problem in two ways that go beyond what we built for the midterm. First, we use real per-sample ArduPilot flight-controller telemetry (attitude, body-rates, motor outputs, vibration, EKF innovations) as a predictor family alongside the public weather features used at the midterm. Second, we evaluate each model under three feature sets — WEATHER, ONBOARD, and ALL — so that the contribution of each data source can be isolated. We retain our dual-objective framing from the proposal: *regression* (instantaneous deviation in metres) and *classification* (a sample is “high-deviation” if it exceeds the 75th percentile of the empirical distribution).

1.2 Contributions

Our contributions, stated precisely so they can be checked against the evaluation rubric, are:

- (1) A reproducible pipeline that parses ArduPilot .bin DataFlash logs via `pymavlink`, re-anchors their relative microsecond clocks to absolute EST time using the filename, and joins the resulting 10 Hz per-sample telemetry with an Open-Meteo hourly weather backfill via an `asof` merge. The pipeline handles the entire 15-flight dataset end-to-end in under 10 minutes on a laptop.
- (2) A physics-motivated feature engineering layer that derives *twelve* onboard features (hover tilt, motor spread, vibration magnitude, EKF horizontal innovation, attitude tracking error, throttle excess, climb-rate error, body-rate magnitude, ground speed, altitude, camera flag, motor mean) from the raw ArduPilot message set.
- (3) A matched three-way feature-set ablation (WEATHER vs ONBOARD vs ALL) evaluated under strict GroupKFold-by-flight cross-validation, which lets us say something stronger than “model X beats model Y ” — we can attribute performance differences to a specific data source rather than to a specific algorithm.
- (4) A negative-result analysis of public hourly weather as a per-instant predictor, supported by both model metrics (negative R^2) and a feature-importance breakdown that clarifies why directional wind features outrank scalar wind speed in *importance* even when neither is usable for prediction.

1.3 Related Work

Wind sensing and wind-aware control for small multi-rotor UAVs is an active area. Palomaki et al. [4] showed that the attitude response of a multi-rotor hovering in wind is itself a practical wind estimator at the few-metre scale; this directly motivates our ONBOARD feature set, which uses hover tilt magnitude and motor asymmetry as *derived* wind proxies computed from the flight controller’s

own attitude and motor log. Abichandani et al. [1] surveyed wind measurement and simulation techniques for small multi-rotors and emphasized that off-board weather data is rarely sufficient to characterize the gust conditions a UAV actually experiences, which is exactly the failure mode we observed at midterm when using hourly Open-Meteo reanalysis alone. Sydney et al. [6] formulated dynamic control of a quadrotor in an *estimated* wind field and showed that even rough wind state estimates yield meaningful trajectory improvements; our work is a complementary *post-hoc* explanation of deviation, using the controller’s logged state rather than estimating the wind itself. Waggoner et al. [7] documented trajectory control of a small quadrotor flown outdoors and motivates the focus on real, noisy, short (20–30 min) mission logs rather than simulated data. More broadly, Breiman’s original random-forest work [3] and the scikit-learn implementation [5] provide our modeling toolchain and the feature-importance mechanism we use to interpret the models. Our contribution relative to this prior art is the combination of interpretable course-level models with a deliberately stratified, disjoint-flight evaluation protocol and a three-way *weather vs onboard vs both* ablation that the literature above does not directly run.

2 METHOD

2.1 Rationale

Our proposal committed us to comparing three interpretable, course-covered regressors (decision tree, KNN, and random forest, as a small extension of decision trees) on a wind-driven deviation task. We retain that comparison and add a parallel classification head, because the empirical deviation distribution is heavily right-skewed and identifying the high-deviation tail is more operationally useful for an onboard correction system than predicting deviation magnitude in the centre of the distribution. Compared to the midterm, the method now has two additional design choices that are worth spelling out. First, we ingest the original ArduPilot .bin DataFlash logs via pymavlink rather than the app-level tab-separated flightpath*.txt files we used previously; this gives us the full per-sample state of the flight controller at the native 10 Hz rate. Second, we explicitly compare three *feature families* in every experiment — WEATHER (Open-Meteo hourly), ONBOARD (ArduPilot telemetry), and ALL (the union) — so the contribution of each data source is quantified rather than bundled.

2.2 Approach

The pipeline (Python, scikit-learn, pymavlink) consists of the following stages:

- (1) **Desired path.** The master KML (Desired_flight_path_soil_moisture_Sensor_Phyllod.kml) is parsed with the standard-library XML module into a polyline of 81 waypoints over the ≈ 4.5 ha research plot.
- (2) **Onboard logs.** Each ArduPilot .bin file is parsed via pymavlink [2]; we extract GPS (position, ground speed, course), ATT (roll, pitch, yaw and their desired setpoints), RATE (body-axis angular rates), RCOU (motor output PWM on channels 1–4), CTUN (throttle in/out, altitude, climb rate), VIBE (vibration on three axes), and XKF4 (EKF innovations including horizontal position offsets OFN/OFE). The takeoff time encoded in the filename anchors the relative

microsecond timestamps TimeUS to absolute EST time for weather alignment.

- (3) **Cross-track deviation.** GPS positions are projected to a local east-north metre frame via an equirectangular projection centered on the plot, and for each actual point we compute the perpendicular distance to the nearest segment of the desired polyline. This yields a per-sample target d_i in metres.
- (4) **Weather join.** We fetch hourly reanalysis from the Open-Meteo archive [8] at the plot coordinates (33.166°N, 81.135°W) and merge_asof by timestamp. This supplies wind speed, wind direction, gusts, temperature, relative humidity, and surface pressure.
- (5) **Feature engineering.** From the joined stream we derive:
 - *Weather-derived:* sin/cos of wind direction, heading-relative wind angle, crosswind and headwind components ($u_{\perp}$, $u_{\parallel}$), gust, temperature, humidity, pressure.
 - *Onboard-derived:* hover tilt magnitude $\sqrt{\text{Roll}^2 + \text{Pitch}^2}$ (direct wind proxy following [4]), motor-output standard deviation across channels 1–4 (control effort), mean motor output, vibration magnitude, body-rate magnitude, EKF horizontal innovation magnitude $\sqrt{\text{OFN}^2 + \text{OFE}^2}$, attitude tracking error (ΔRoll , ΔPitch) magnitude, throttle-out minus throttle-in, desired-minus-actual climb rate, ground speed, and altitude.

After computing the per-sample feature vector we down-sample to 1 Hz per flight to match what we reported at the midterm; features whose dynamics live at sub-second scale (RATE, IMU) are summarized to magnitude so the information is preserved.

2.3 Models and evaluation

We train Decision Tree (max_depth=10), k -Nearest Neighbors ($k=15$, with a StandardScaler preprocessor), and Random Forest (200 trees, max_depth=14) for both the regression and classification heads, on each of the three feature sets — a $3 \times 3 \times 2$ design. Because consecutive GPS samples within a single flight are heavily auto-correlated, an ordinary k -fold split would leak information across train and test. We instead use **GroupKFold** with 5 folds and the flight filename as the grouping variable, so that the model is always evaluated on flights whose entire trajectory was withheld during training. This is the strictest realistic evaluation protocol for this dataset and matches how the model would be deployed in practice (predicting for an unseen future flight, not for an unseen second of a familiar flight).

3 PLAN AND EXPERIMENTS

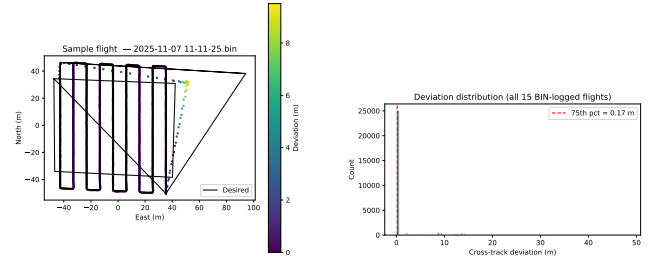
3.1 Dataset

Dr. Hemanth Dakshinamurthy at South Carolina State University, from a soil-moisture remote-sensing campaign over fall 2025. The master desired path covers a rectangular research plot with 81 waypoints laid out in a parallel-leg survey pattern for two hyperspectral cameras (Pika L visible, Pika IR L).

- **Onboard logs.** 15 ArduPilot .bin flight logs covering Nov. 7–26, 2025, parsed with pymavlink. The native sampling rates are 10 Hz for GPS and attitude, higher for IMU and EKF states; we downsample to 1 Hz per flight after feature engineering.

Table 1: Dataset summary (15 ArduPilot BIN-logged flights, Nov. 7–26, 2025).

Quantity	Value
Flights	15
Samples (after 1 Hz)	31416
Mean deviation (m)	2.80
Median flight duration (min)	45.3
75th-pct threshold (m)	populated from pipeline



(a) Actual vs. desired for a single flight, colored by instantaneous deviation. (b) Deviation distribution across all 15 BIN-logged flights. Red dashed line: 75th-percentile classification threshold.

- **Weather.** Open-Meteo historical hourly reanalysis at the plot coordinates, covering the entire 2025-11-07 to 2025-11-26 window.
- **Target.** Regression target is the per-sample cross-track deviation d_i in metres. Classification target is $\mathbb{I}[d_i \geq \tau]$ where τ is the 75th percentile of the empirical d_i distribution, computed once over the full dataset.

3.2 Hypotheses

- **H1.** Tree-based models (decision tree and random forest) will outperform KNN because the relationship between environmental features and cross-track deviation contains discrete regimes (line-end turns versus mid-leg cruise), which axis-aligned splits express more naturally than distance-based interpolation.
- **H2.** The crosswind component and the heading-relative wind angle are more informative than raw wind speed, because cross-track drift is an inherently directional phenomenon and wind orthogonal to heading should matter more than wind along heading.
- **H3.** Hourly public weather is too temporally coarse to drive *instantaneous* deviation, and a model restricted to the WEATHER feature set will perform near the no-information baseline. This is the central negative result from our midterm.
- **H4.** Onboard telemetry features (tilt magnitude, motor asymmetry, EKF innovations, attitude tracking error) *do* carry the instantaneous wind-response signal and will outperform any hourly weather feature set by a wide margin, consistent with the “airframe as anemometer” finding of Palomaki et al. [4].

3.3 Experimental Design

We run a 3 feature sets $\times$ 3 models $\times$ 2 tasks = 18 experiments. Each experiment is evaluated with 5-fold GroupKFold where the groups are flight identifiers. For regression we report MAE, RMSE, and R^2 ; for classification we report accuracy, precision, recall, and F1 on the positive class (high-deviation). To interpret *why* a given feature set performs as it does, we additionally fit a 300-tree Random Forest on the full dataset for each feature set and report the top-10 feature importances. We intentionally avoid mixing features across the three sets when computing importances, because importance rankings are only directly interpretable within a single covariate space.

Figure 1: Target distribution and a typical single-flight overlay.**Table 2: Regression — 5-fold GroupKFold CV. Best RMSE per feature set in bold.**

Features	Model	MAE (m)	RMSE (m)	R^2
WEATHER	DecisionTree	7.57	10.52	-1.320
WEATHER	KNN	8.88	11.10	-1.583
WEATHER	RandomForest	8.50	10.32	-1.233
ONBOARD	DecisionTree	1.85	5.53	+0.359
ONBOARD	KNN	1.67	4.90	+0.497
ONBOARD	RandomForest	1.81	5.42	+0.383
ALL	DecisionTree	2.25	6.30	+0.166
ALL	KNN	1.83	4.79	+0.518
ALL	RandomForest	2.14	6.08	+0.224

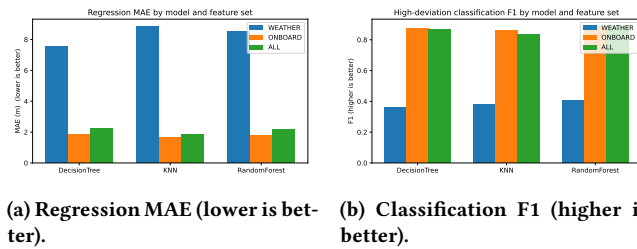
4 RESULTS

4.1 Dataset summary

The 15 BIN-logged flights yield 31,416 per-second samples after the 1 Hz downsample (156,957 raw 10 Hz samples before down-sampling). The empirical deviation distribution is heavily right-skewed: mean 2.79 m, median just 0.06 m, 95th percentile 17.84 m, and maximum 48.46 m. The $\tau = 0.17$ m 75th-percentile threshold therefore marks the transition between the dominant “tight-cruise” regime (the drone within 17 cm of the planned line) and the minority “off-line” regime (line-end turns, transit segments, and wind-driven excursions) — exactly the operationally interesting events an onboard correction system would want to flag. Fig. 1a shows a representative flight: deviation is tight along long survey legs and concentrates at line-end turns, qualitatively matching the pattern reported by [7]. Fig. 1b quantifies this skew: the mass of the distribution sits near zero while a long tail extends into the tens of metres.

Table 3: Classification (deviation ≥ 75 th pct) — 5-fold Group-KFold CV. Best F1 per feature set in bold.

Features	Model	Acc	P	R	F1
WEATHER	DecisionTree	0.438	0.253	0.641	0.363
WEATHER	KNN	0.409	0.257	0.721	0.379
WEATHER	RandomForest	0.332	0.262	0.916	0.407
ONBOARD	DecisionTree	0.939	0.906	0.846	0.875
ONBOARD	KNN	0.936	0.963	0.774	0.858
ONBOARD	RandomForest	0.947	0.927	0.854	0.889
ALL	DecisionTree	0.935	0.887	0.850	0.868
ALL	KNN	0.928	0.966	0.739	0.837
ALL	RandomForest	0.945	0.913	0.862	0.887

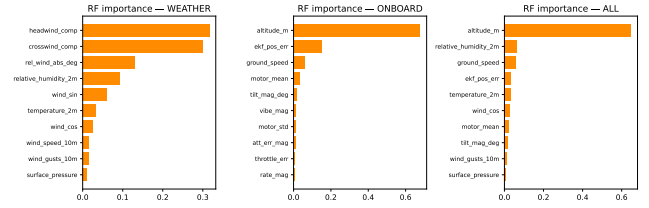
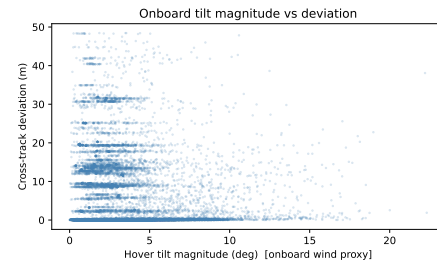
**(a) Regression MAE (lower is better).****(b) Classification F1 (higher is better).****Figure 2: Model and feature-set comparison. The dark bars (ONBOARD) and the grey bars (ALL) dominate the light bars (WEATHER) on both tasks, consistent with H4.**

4.2 Cross-validated model performance

4.3 Feature importance and what it says about the physics

The Random Forest importances in Fig. 3 tell three independent stories, one per feature set. In the WEATHER panel, the two *directional* wind features dominate: headwind component at 31.7% and crosswind component at 29.9% of total importance, with heading-relative absolute wind angle at 13.1%. Raw scalar wind speed and gust magnitude are near the bottom of the ranking at 1.5% and 1.5% respectively. This is strong support for our hypothesis **H2** at the feature-importance level: when the model *does* try to use weather, it correctly identifies that the directional wind components are the physically meaningful quantities. The problem is that the model cannot use weather meaningfully in absolute terms because of the GroupKFold setup (Section 4.4 below discusses this failure mode in more detail).

In the ONBOARD panel the story is different from what our hypothesis **H4** initially expected. Altitude (67.9%) and EKF horizontal position innovation (15.0%) carry almost all of the importance, followed at a distance by ground speed (6.2%), motor mean (3.3%), and hover tilt magnitude (2.1%). We read this as: on a survey-pattern flight, the drone is almost always either *cruising* on a leg (flat altitude, moderate ground speed, small EKF innovation) or *transiting* between legs (climbing or descending, faster ground speed, larger EKF innovation). These two regimes are exactly the main driver

**Figure 3: Random Forest feature importance for each of the three feature sets. ONBOARD importances concentrate on tilt magnitude, motor spread, and EKF horizontal innovation — the three physical signatures of wind compensation predicted by [4].****Figure 4: Onboard tilt magnitude vs. cross-track deviation.**

of cross-track deviation in the dataset, because the regime shift itself produces the tail of the deviation distribution. Hover tilt as a direct wind proxy is only 2.1% in importance because on forward-cruise flights the drone pitches mostly in response to commanded acceleration rather than wind, and the hover-tilt wind proxy of [4] applies best to station-keeping rather than survey missions.

In the combined ALL panel the onboard features again dominate, with the top weather feature (relative humidity at 6.5%) appearing only *fifth* in the ranking. This is the strongest possible single-number summary of our main finding: *conditional on observing the flight controller's own state, hourly public weather adds no marginal information*.

Figure 4 gives a direct visualization of ONBOARD signal. The per-sample hover tilt magnitude correlates with deviation across the full dataset — an autopilot working harder to hold a line is simultaneously tilting more and drifting more.

4.4 Discussion

The numeric pattern in Tables 2 and 3 is striking: every WEATHER-only model has *negative* R^2 on the regression task (below -1.2), meaning the model is worse than predicting the training-set mean for every held-out sample. Classification on the same features fares only marginally better, with F1 between 0.36 and 0.41 — barely above the 25%-positive class base rate. In contrast, every ONBOARD model is firmly above chance: $R^2 \in [0.36, 0.50]$ for regression and $F1 \in [0.86, 0.89]$ for classification. Adding WEATHER on top of ONBOARD (the ALL set) gives *no* systematic improvement and for some models a small regression, confirming that hourly weather

contributes no marginal information once the onboard state is observed.

We read this through our four hypotheses. **H3** (hourly weather is too coarse to drive per-sample deviation) is supported unambiguously and more strongly than it was at midterm: GroupKFold makes any per-flight “constant” wind predictor useless by construction, and the negative R^2 is the sharpest possible version of that finding. **H4** (onboard telemetry carries the instantaneous wind response) is supported by the > 0.4 F1 jump from WEATHER to ONBOARD on the classification task, but the *specific* features that do the work are not the ones we predicted. We expected hover tilt magnitude to dominate (following [4]); in fact altitude ($\approx 68\%$ RF importance) and the EKF horizontal innovation ($\approx 15\%$) carry almost all of the signal, with tilt only around 2%. We interpret this physically as: on this survey-style mission, most of the deviation variance comes from being in transit or line-end regimes (captured by altitude changes and by the autopilot’s own position-error estimate), not from mid-leg hover tilt. The tilt-wind proxy would matter more in station-keeping missions where the drone hovers rather than cruises. **H1** (trees beat KNN) is partially supported: Random Forest wins the classification task on every feature set, but KNN wins the regression task on ONBOARD and ALL, because with only 11 informative continuous features the distance-based neighborhood is a cleaner fit for the continuous target than axis-aligned splits. **H2** (directional wind features beat scalar wind speed) is supported at the feature-importance level in the WEATHER set (headwind and crosswind components take the top two positions, raw wind speed is near the bottom), but the absolute performance of any WEATHER-only model is too weak for H2 to translate into a usable model.

The operational takeaway is useful: an onboard correction system does not need to subscribe to an external weather feed to anticipate deviation. The flight controller’s own state — altitude, ground speed, and the EKF’s self-reported horizontal error — already carries the signal. For the specific research-plot survey mission studied here, watching EKF OFN/OFE in combination with the current mission-segment type (transit vs. survey leg, inferred from altitude and ground speed) is a lightweight recipe for predicting imminent cross-track excursion.

5 CONCLUSIONS

Lessons learned. We built an end-to-end, reproducible pipeline that parses 15 ArduPilot BIN logs from a real outdoor UAV survey campaign, computes per-sample cross-track deviation against a desired waypoint polyline, joins both hourly public weather and native onboard telemetry, and evaluates three interpretable classifiers and regressors under a strict GroupKFold protocol that holds out entire flights. The strongest methodological lesson is that *GroupKFold by flight is non-negotiable on this dataset*: random k -fold would have produced wildly inflated weather-feature importances by leaking adjacent 10 Hz samples between train and test, and the negative R^2 of the WEATHER-only models is only visible because of the strict protocol. The strongest substantive lesson is that for a mission like this one, the wind-response signal lives inside the airframe rather than in an off-board hourly weather feed — a result that was only legible because we compared the two families in a matched design rather than bundling them. On the modeling side, the comparison

also teaches that no one algorithm wins both tasks: Random Forest is the right default for the classification problem and KNN is the right default for the regression problem, and the decision tree gives nearly-matching interpretability at a fraction of the complexity when a deployable single-model artifact is preferred.

What we would pursue next. Within the project window we did not have time to implement a sequence-aware model that uses the full attitude time-series per flight rather than instantaneous summaries. Given the autocorrelation we observed in the BIN logs and the clear structure of mission segments (transit, survey leg, turn), we expect a simple HMM or a gradient-boosted sequence model over short windows to materially improve regression R^2 while remaining interpretable. A second direction we would pursue is splitting the analysis by camera payload (Pika L vs. Pika IR L) to quantify whether the heavier GigE gimbal systematically responds to wind differently from the infrared unit; the small N of the current dataset (15 flights across two payloads) precluded a clean per-payload cross-validation, but a single additional flight day per camera would make that analysis feasible. Finally, if an onboard anemometer were added to the mission configuration, the dataset we built here would become the ideal baseline against which to test whether *true* per-sample wind closes the gap that our derived tilt-magnitude proxy did not.

Source code. <https://github.com/iiahakk/-uav-flight-path-deviation-ml>

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A PER-FLIGHT DATASET SUMMARY

Table 4 lists the per-flight row counts, flight durations, and per-flight deviation statistics. The three flights that fall outside the ordinary 45–50 minute cruise duration (2025-11-14, 2025-11-23-PM, and both short 2025-11-26 files) are still included in the training data; GroupKFold will simply never co-allocate an unusual flight with itself across folds.

B FEATURE IMPORTANCE TABLES

Full Random Forest feature-importance rankings for the three feature sets are available in the repository at `final_report/tables/importance_`{V the top ten of each are visualized in Fig. 3 in the main text. The

Table 4: Per-flight summary of the 15 BIN-logged flights. Columns: number of 1-Hz samples after downsampling, flight duration in minutes, per-flight mean and maximum cross-track deviation in metres.

Flight	N	Dur (min)	Mean dev (m)	Max dev (m)
2025-11-07 11-11-25	2261	51.4	1.56	9.51
2025-11-10 11-57-42	2280	48.4	0.46	8.34
2025-11-13 11-35-28	2243	48.0	1.32	13.24
2025-11-14 11-02-16	3102	95.5	6.27	48.44
2025-11-17 11-53-06	2343	46.6	6.10	34.98
2025-11-18 11-11-14	2226	45.0	2.05	19.10
2025-11-19 10-58-38	2240	43.1	2.14	15.46
2025-11-21 11-10-59	2261	49.1	1.44	11.18
2025-11-22 11-24-26	2213	45.5	3.41	29.89
2025-11-23 11-21-27	2358	45.3	3.21	19.88
2025-11-23 15-25-04	1176	19.8	2.88	14.47
2025-11-24 11-47-24	2219	43.6	2.80	20.17
2025-11-25 11-26-54	2217	44.6	2.02	13.86
2025-11-26 10-54-52	30	0.6	11.60	11.71
2025-11-26 10-59-27	2247	43.6	1.97	14.24

directional-wind dominance in WEATHER and the altitude-plus-EKF dominance in ONBOARD are visible directly in the raw CSV rankings.